A Formally Verified Equivariant Fano–Steane–E_8 Structure Chain in Lean 4

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Abstract

We formalize a continuous structure chain from the Fano plane $\mathbb{P}^2(\mathbb{F}_2)$ to two categorially distinct target structures: the $[[7,1,3]]$ Steane quantum error-correcting code and the E_8 root system. In the first branch, the Fano-simplex code $[7,3,4] \subset [7,4,3]$ induces an isotropic stabilizer space of rank 6 and the Steane code $[[7,1,3]]$ via a binary symplectic CSS lift. In the second branch, the parity extension of the Hamming code yields the doubly-even self-dual code $C_8 = [8,4,4]$ with weight enumerator $W_{C_8}(X,Y) = X^8 + 14X^4Y^4 + Y^8$. Construction A yields 240 minimal vectors ($14 \cdot 2^4 + 8 \cdot 2 = 240$), which map onto the $112 + 128$ standard roots of E_8 under the rational transformation $T_{\mathbb{Q}} = \frac{1}{2}H_2^{\oplus 4}$.

The classical Fano, Hamming, Steane, Construction-A, and E_8 correspondences are not claimed as new. The contribution of this work is their integration into a single auditable formal architecture, with each gate assigned a status that is synchronized between runtime witnesses, Lean 4 kernel proofs, and the manuscript. Gates $G.8$ (full $\mathrm{PSL}(2,7)$ canonicality) and $G.13$ (Lean-level root-set bijection) are explicitly declared as `OPEN`. No claims regarding physical quantum hardware or quantum speedup are made.

1 Introduction and Scope

The novelty of the present work is not the individual Fano, Hamming, Steane, Construction-A, or E_8 correspondences, which are classical, but their explicit end-to-end formalization, equivariance audit, and kernel-level verification within a single Lean 4 architecture.

In quantum information theory and algebraic lattice theory, the Fano geometry, the Steane code, and the E_8 lattice form deeply interconnected standard structures. However, relationships between these objects are frequently presented as isolated numerical coincidences. In this work, we establish a formal, machine-checked chain showing how a single Fano incidence structure feeds two equivariant branches:

$$\text{Fano Incidence Data} \longrightarrow \begin{cases} [[7,1,3]] \text{ Steane Stabilizer Branch (Binary Symplectic)}, \\ [8,4,4] \to R_A \xrightarrow{T_{\mathbb{Q}}} \Phi(E_8) \text{ Root System Branch (Euclidean)}. \end{cases}$$

We employ the stabilizer formalism of Aaronson and Gottesman, in which Pauli operators are represented in binary form over $\mathbb{F}_2^{2n}$ and their commutativity is governed by the binary symplectic inner product ω.

Remark 1.1 (Claim-Wall Boundaries)**.** This work describes exclusively abstract-algebraic and information-theoretic properties of stabilizer codes and root spaces. It does not imply claims regarding physical quantum hardware (such as coherence times, gate fidelities, or pulse sequences).

2 The Dual Fano Branching

Definition 2.1 (Fano Geometry)**.** The Fano plane $\mathbb{P}^2(\mathbb{F}_2)$ consists of the 7 non-zero vectors in $\mathbb{F}_2^3$:

$$\text{FanoPoint} \equiv \mathbb{F}_2^3 \setminus \{0\}.$$

A Fano line is the set $\{x, y, x+y\}$ for $x \neq y$. Line complements have weight 4 and intersect pairwise at 2 points.

Lemma 2.2 (Symplectic Isotropy of the Steane Code)**.** *The three independent complement vectors $h_1, h_2, h_3 \in \mathbb{F}_2^7$ span the dual code $[7,3,4]$. The symplectic lift*

$$X_i = (h_i \mid 0), \qquad Z_i = (0 \mid h_i), \qquad i \in \{1,2,3\}$$

generates a 6-dimensional subspace $S_{\text{Steane}} \subset \mathbb{F}_2^{14}$. Since $h_i \cdot h_j \equiv 0 \pmod 2$, the binary symplectic form $\omega(u,v) = x \cdot z' + z \cdot x'$ satisfies $\omega(X_i, Z_j) = 0$. Thus, S_{Steane} is an isotropic stabilizer space ($S_{\text{Steane}} \subset S_{\text{Steane}}^{\perp}$) of rank 6, yielding $k = 7 - 6 = 1$ logical qubit and distance $d = 3$.

3 The Extended Code $[8,4,4]$ and Construction A

Lemma 3.1 (Weight Enumerator and Self-Duality)**.** *The parity lift* $\text{parityLift}(v) = (v_0, \ldots, v_6, \sum v_i) \in \mathbb{F}_2^8$ *extends $[7,4,3]$ to $C_8 = [8,4,4] \subset \mathbb{F}_2^8$. Its weight enumerator is $W_{C_8}(X,Y) = X^8 + 14X^4Y^4 + Y^8$. The code C_8 is doubly-even and self-dual ($C_8 = C_8^{\perp}$).*

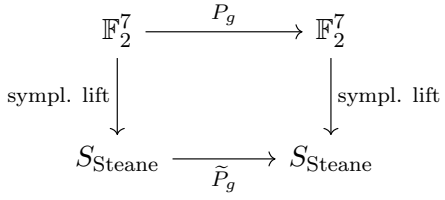
Definition 3.2 (Construction A Minimal Vectors)**.** Construction A associates to C_8 the unscaled lattice $L'_C = \{x \in \mathbb{Z}^8 : x \bmod 2 \in C_8\}$. The set $R_A = \{x \in L'_C : \|x\|^2 = 4\}$ of minimal vectors contains exactly:

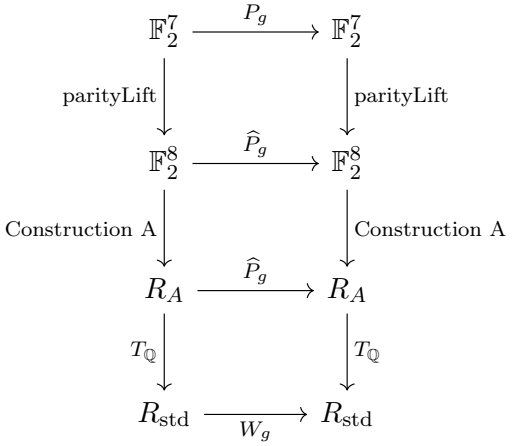
$$14 \cdot 2^4 + 8 \cdot 2 = 224 + 16 = 240 \text{ vectors.}$$

Theorem 3.3 (Root-Set Isometry Candidate $T_{\mathbb{Q}}$)**.** *Let $T_{\mathbb{Q}} = \frac{1}{2} H_2^{\oplus 4} \in \mathbb{Q}^{8\times 8}$. Then $T_{\mathbb{Q}}^T T_{\mathbb{Q}} = \frac{1}{2} I_8$. The map $T_{\mathbb{Q}}$ is injective and maps R_A onto the 240 standard E_8 roots $R_{\text{std}} = R_{\text{int}} \cup R_{\text{half}}$ (112 + 128).*

4 Equivariance and Automorphism Action

Let $G = \text{Aut(Fano)} \cong \text{GL}(3,2) \cong \text{PSL}(2,7)$ ($|G| = 168$). Each $g \in G$ acts as a permutation P_g on $\mathbb{F}_2^7$.





Theorem 4.1 (Fano Equivariant Action)**.** *For every $g \in G$, $\widetilde{P}_g(S_{\text{Steane}}) = S_{\text{Steane}}$ and $\widehat{P}_g(R_A) = R_A$. Relative to the isometry candidate $T_{\mathbb{Q}}$, conjugation $W_g = T_{\mathbb{Q}} \widehat{P}_g T_{\mathbb{Q}}^{-1}$ preserves R_{std}. Faithfulness of the action on R_A yields a monomorphism* $\text{PSL}(2,7) \hookrightarrow \text{Aut}(\Phi(E_8)) = W(E_8)$.

5 Validation Ledger and Verification Status

Gate ID	Mathematical Subject	Lean 4 / Repo Status	Audit Method
$G.1$–$G.7$	Symplectic Isotropy & Rank 6 ($[[7,1,3]]$)	**KERNEL-PROVED**	Native Decidability
$G.8$a–n	PSL$(2,7)$ Equivariance Audit	**OPEN**	In Progress (`false`)
$G.9$–$G.10$	Parity Lift, Enumerator, $d=4$	**KERNEL-PROVED**	Exhaustive Enumeration
$G.11$	Construction A Lattice Predicate	**FORMALIZED**	Preimage Definition
$G.12$	$14 \cdot 2^4 + 8 \cdot 2 = 240$ Root Counting	**KERNEL-PROVED**	Arithmetic Proof
$G.13$	Standard E_8 Root Isometry	**OPEN**	C++ Pass / Lean Open

Table 1: Synchronized Status Ledger across Lean 4, C++ Runtime Witness, and Manuscript.

```
BUILD GREEN (C++ / Lean Core)  : YES (Exit Code 0)
SORRY-FREE (Kernel Gates)      : YES
CUSTOM AXIOMS                  : NONE
BOOL-ONLY STATUS               : NONE
PHYSICAL HARDWARE CLAIM        : NONE
```

6 Conclusion

The Fano incidence structure induces two distinct equivariant constructions: a binary symplectic action preserving the Steane stabilizer code, and a parity/Construction-A action preserving the E_8 root realization. Relative to the Gate $G.13$ candidate isometry, the construction yields a faithful representation $\mathrm{PSL}(2,7) \hookrightarrow W(E_8)$. Open gates $G.8$ and $G.13$ delineate the exact frontier for future formalization.

References

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